

GenAI Ops: Automated Command Risk Assessment Matrix

Duration : 90 Min.

Scenario

Build a Python security filter ('command_guard.py') that evaluates AI-suggested bash commands against a production safety matrix before execution.

Target File Location & Creation

File to Create/Update: student_workspace/command_guard.py **Input Resource File to Inspect:** student_workspace/untrusted_bash_commands.txt

Task Objectives

Perform the following actions inside the student_workspace directory: - Inspect untrusted_bash_commands.txt in student_workspace/. - Create or update student_workspace/command_guard.py and populate it with valid analysis results.

Instructions to Perform the Task

1. When your workspace loads in **VS Code**, use the **Explorer** panel on the left to locate your files.
2. Open and inspect untrusted_bash_commands.txt inside student_workspace/.
3. Create or open student_workspace/command_guard.py in student_workspace/ and record your answers.
4. Save your changes (Ctrl + S or Cmd + S).
5. Open the built-in terminal by clicking **Terminal > New Terminal** from the top menu.
6. Verify your progress by running python run.py locally in the terminal.

Validation

Once you have saved your files and verified your progress, return to the platform dashboard and click the **"Run Test" / "Verify"** button. This will automatically evaluate your changes and generate your score!

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	command_guard.py is syntax-valid Python	3 Marks
TC2	Flags destructive commands (rm -rf, dd, mkfs)	3 Marks

TC3	Flags privileged escalation (sudo su, chmod 777)	3 Marks
TC4	Allows read-only diagnostic commands (cat, grep, ps)	3 Marks
TC5	Assigns risk score (LOW, MEDIUM, HIGH, CRITICAL)	2 Marks
TC6	Outputs risk_assessment.json	2 Marks
TC7	Includes remediation recommendation	2 Marks
TC8	Handles empty/malformed command input gracefully	2 Marks

Total Score: 20 Marks

Important Notes

- This is an auto-evaluated question. Ensure all code edits are properly saved and the 'run.py' checks pass before submission.

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